

Motilal Nehru National Institute of Technology Allahabad
Department of Computer Science and Engineering

Introduction to Artificial Intelligence and Machine Learning (CSN12601)

Lab Manual: Assignment-4

Lab Title: Programs on Decision and Loop Control Statements

Objective:

1. To understand and implement basic mathematical computations in Python, such as finding the LCM and creating multiplication tables.
2. To practice string manipulation techniques, including checking for palindromes, removing punctuations, and counting vowels.
3. To work with file handling in Python, including reading from a file and analyzing its content to count characters, words, and lines.
4. To enhance problem-solving skills using loops, conditionals, and built-in Python functions.

Materials Required:

- Computer with internet access
- Python (latest version)
- Python Integrated Development Environment (IDLE)
- Access to an online Python environment (e.g., Google Colaboratory)

Theory:

Mathematical computations and string manipulations are core areas of programming that involve working with numbers and text data, often using loops, conditionals, and functions.

1. **LCM (Least Common Multiple):** The smallest positive integer that is divisible by two or more numbers. To compute the LCM, one can use the relationship between LCM and GCD: $LCM(a,b) = \frac{|a \times b|}{GCD(a,b)}$.This can be implemented using loops or by leveraging Python's built-in functions.
2. **Multiplication Table:** Creating a multiplication table involves iterating through a range of numbers and displaying their multiples. This can be achieved using for loops to iterate through the desired range.
3. **Palindrome Check:** A palindrome is a string that reads the same backward as forward, like "radar" or "level." Checking for palindromes involves comparing the string with its reverse. This can be efficiently implemented using string slicing or loops.
4. **Removing Punctuations:** Strings often contain punctuation marks that might need to be removed for text processing tasks. This can be achieved using loops to filter out characters or with Python's `str.translate()` and `str.punctuation` from the `string` module.

5. **Counting Vowels:** Vowels (a, e, i, o, u) are fundamental components of strings. Counting vowels involves iterating over each character in the string and checking if it belongs to the set of vowels.
6. **File Handling:** Working with files is crucial for many programming tasks, such as reading data from a file and analyzing its content. Using Python's file handling capabilities (`open`, `read`, and `readlines`), one can count the number of characters, words, and lines in a text file. This involves reading the file content and using string operations like `split()` to break text into words and lines.

List of Programs in Assignment-4

1) Write a program to compute LCM of a number.
2) Write a program to create a multiplication Table.
3) Write a program to print a given string is a palindrome or not.
4) Write a program to remove punctuations from a string.
5) Write a program to print number of vowels present in a string.
6) Write a Python program to compute the number of characters, words and lines in a file.